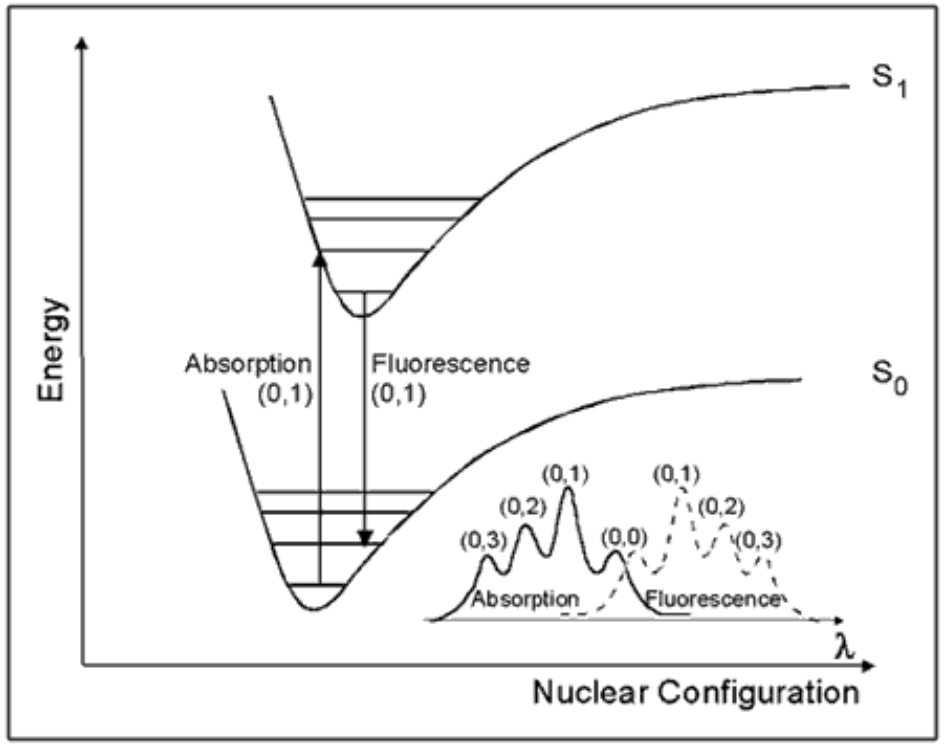
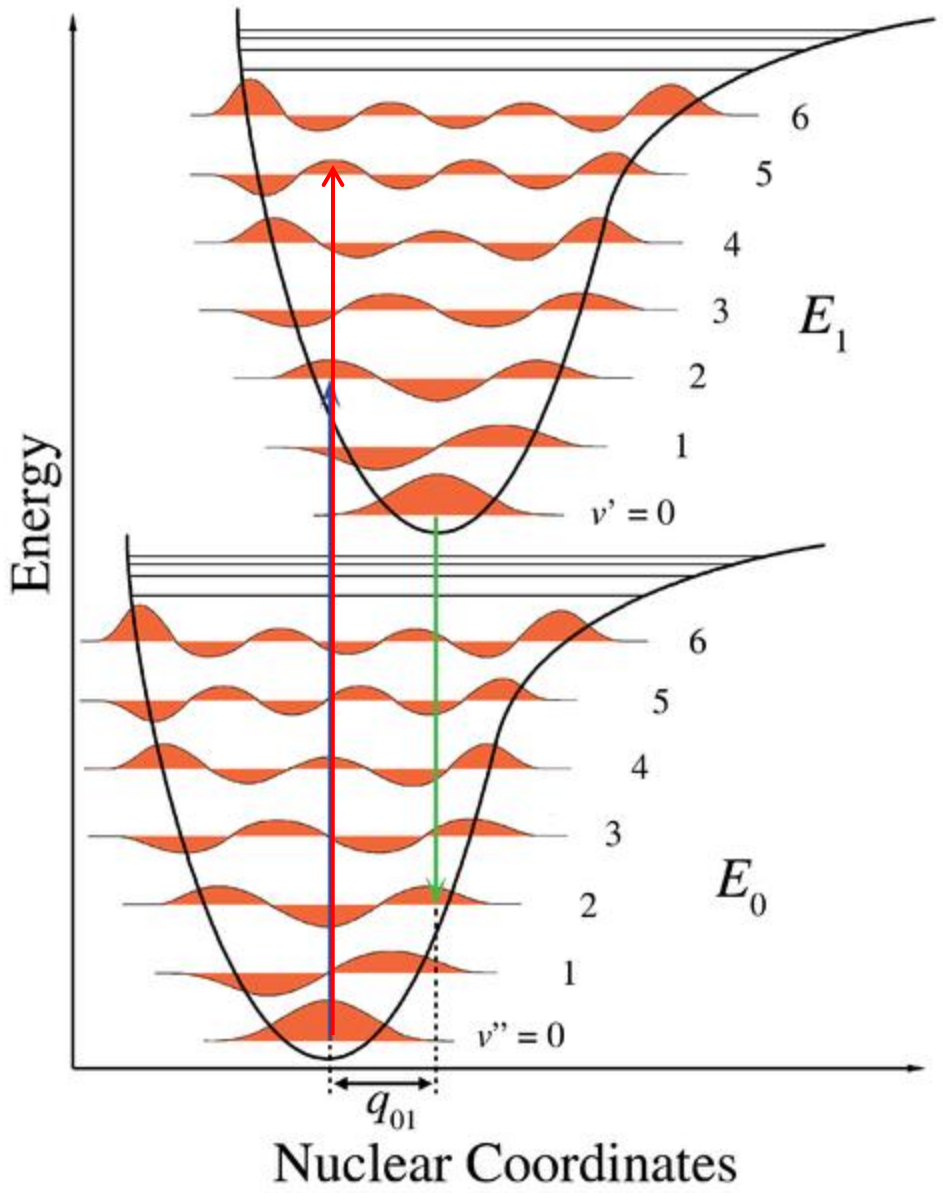
The Franck-Condon Principle

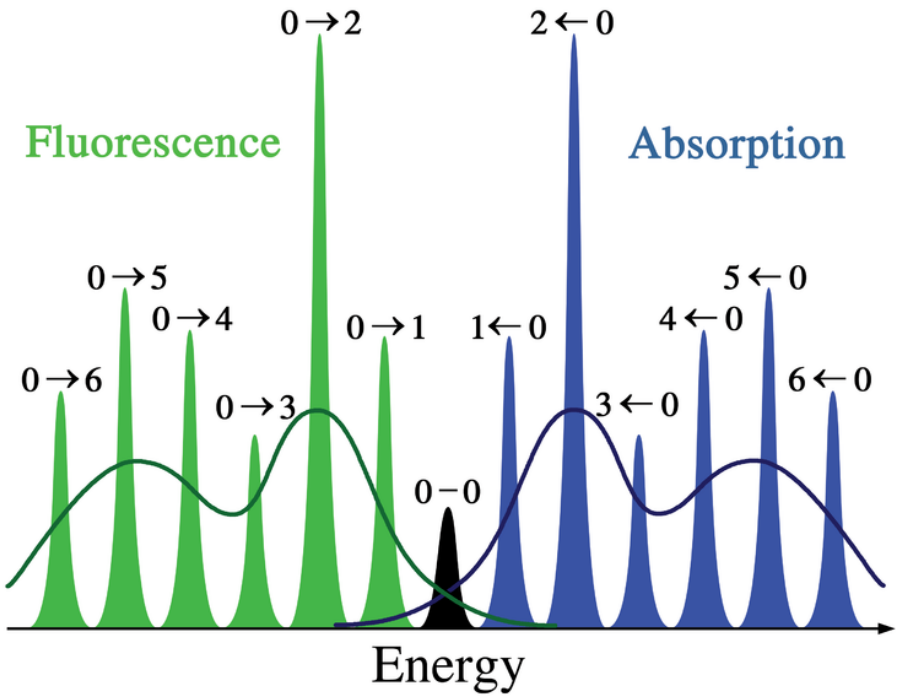
Electronic transitions are so fast (10^{-15} s) in comparison to the nuclear motion (10^{-13} s) that immediately after the transition, the nuclei have nearly the same relative position and momentum as they had just before the transition.

The most probable transitions between different electronic and vibrational levels are those for which the momentum and the position of the nuclei do not change very much.

The most probable transition appears as the most intense absorption line at a frequency given by Bohr relationship.




Energy
6
5
4
3
2
1
v' = 0
E1
6
5
4
3
2
1
v'' = 0
E0
q01
Nuclear Coordinates


Fluorescence
Absorption
0→2
2←0
0→5
0→4
0→1
1←0
5←0
4←0
0→6
0→3
3←0
6←0
0−0
Energy

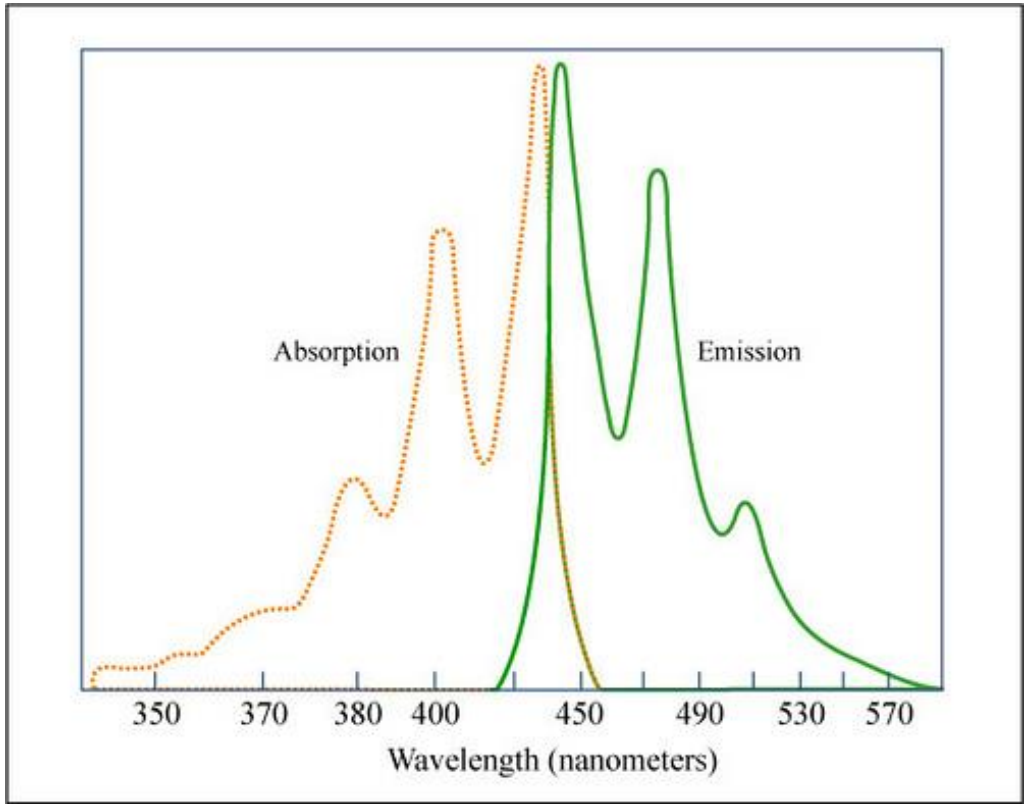
Emission Spectra

The emission of radiation is also governed by Franck-Condon principle.

The molecule comes to stay at the zero vibrational level of the first excited state for about 10^{-8} s for an allowed transition and more for a partially forbidden transition.

If it is permitted to return to the ground state by radiative transition *fluorescence emission* is observed.

If the internuclear geometries are not very different, the fluorescence spectrum appears as a *mirror image* of the corresponding absorption spectrum with a region of overlap.

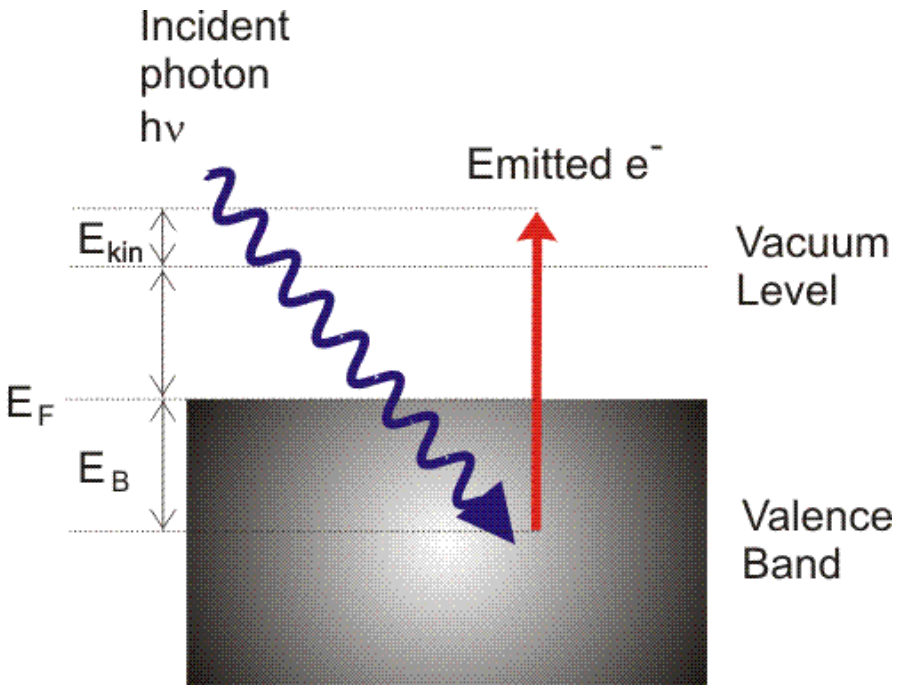


Photoelectron Spectroscopy

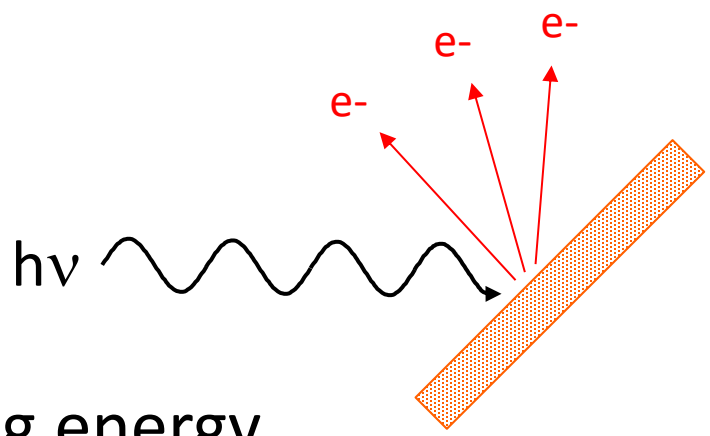
Photoelectron spectroscopy is based on Einstein's photoelectric effect.

A photon can ionize an electron from a molecule if the photon has an energy greater than the energy holding the electron in the molecule.

Any photon energy in excess of that needed for ionization is carried by the outgoing electron in the form of kinetic energy.



Ionization occurs when matter interacts with light of sufficient energy.



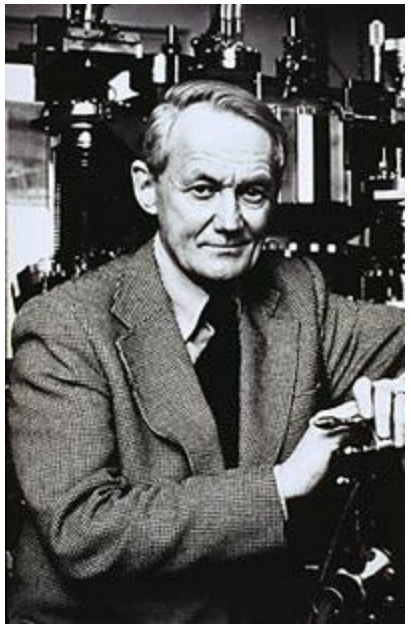
$E_{h\nu}$ = electron kinetic energy + electron binding energy

Ionization is still a transition between states

- Initial State: Neutral (or anion)
- Final State: Atom/Molecule/Cation after an electron is removed, plus the ejected electron
- $M \rightarrow M^{+} + e^{-}$

Kai Seigbahn: Development of X-ray Photoelectron Spectroscopy

C. Nordling E. Sokolowski and K. Siegbahn, *Phys. Rev.* **1957**, *105*, 1676.



Precision Method for Obtaining Absolute Values of Atomic Binding Energies

CARL NORDLING, EVELYN SOKOLOWSKI, AND KAI SIEGBAHN
Department of Physics, University of Uppsala, Uppsala, Sweden
(Received January 10, 1957)

WE have recently developed a precision method of investigating atomic binding energies, which we believe will find application in a variety of problems in atomic and solid state physics. In principle, the method

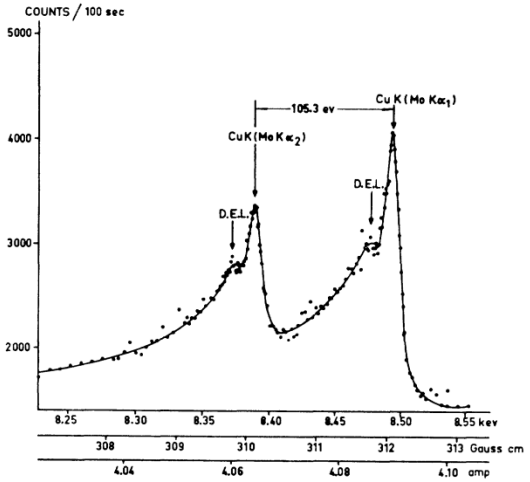


FIG. 1. Lines resulting from photoelectrons expelled from Cu by Mo $K\alpha_1$ and Mo $K\alpha_2$ x-radiation. The satellites marked D.E.L. are interpreted as due to electrons which have suffered a discrete energy loss when scattered in the source.

Nobel Prize in Physics 1981
(His father, Manne Siegbahn, won the Nobel Prize in Physics in 1924 for the development of X-ray spectroscopy)

David Turner: Development of Ultraviolet Photoelectron Spectroscopy

Although a monochromator could provide the monochromatic radiation, the resonance lines of the rare gases form copious and much simpler sources. The helium low pressure discharge is sufficiently monochromatic ($h\nu = 21.3$ eV) for the purpose and gives a convenient spread of photoelectron energy values ($\sim 5 \simeq 15$ eV). In particular the helium resonance line is sufficiently energetic to cause photo-ionization from lower (more tightly bound) levels than the first. Indeed the smaller energy gap $h\nu - I$ would be expected to lead to a higher cross section than for the first ionization process.[2] Such an experiment as has been indicated would therefore be expected to give, in general, a series of electron kinetic energies E_1, E_2, E_3 seen as steps in an R.P.F. curve, or peaks in its differential, corresponding to those energy levels I_1, I_2, $I_3 \cdots$ in the target molecule (or atom) whose term value is less than 21.3 eV. Values of I other than the first are usually difficult to observe by conventional methods in molecules of more than two or three atoms, the method suggested here appears of greater generality.

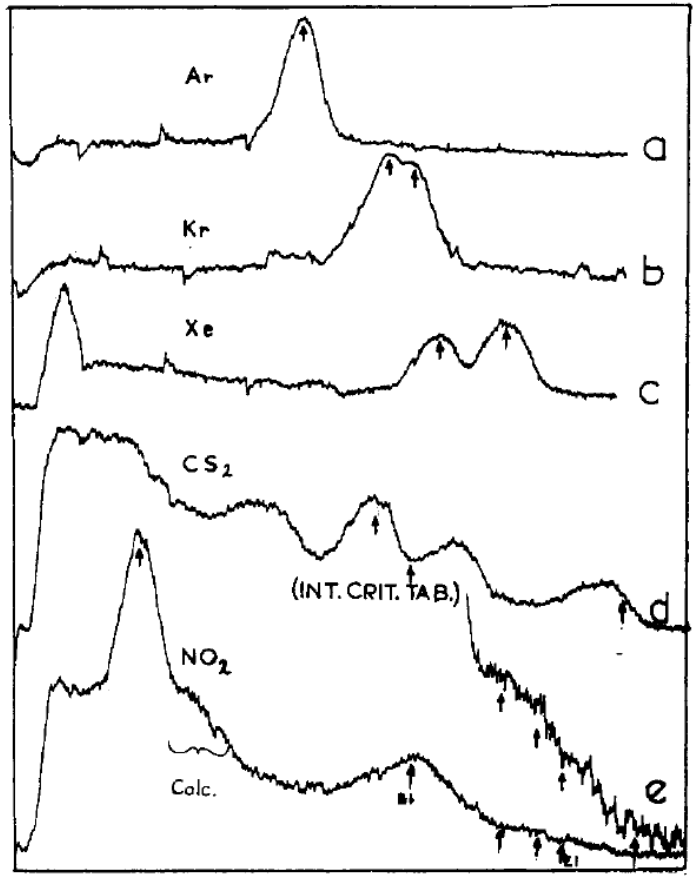
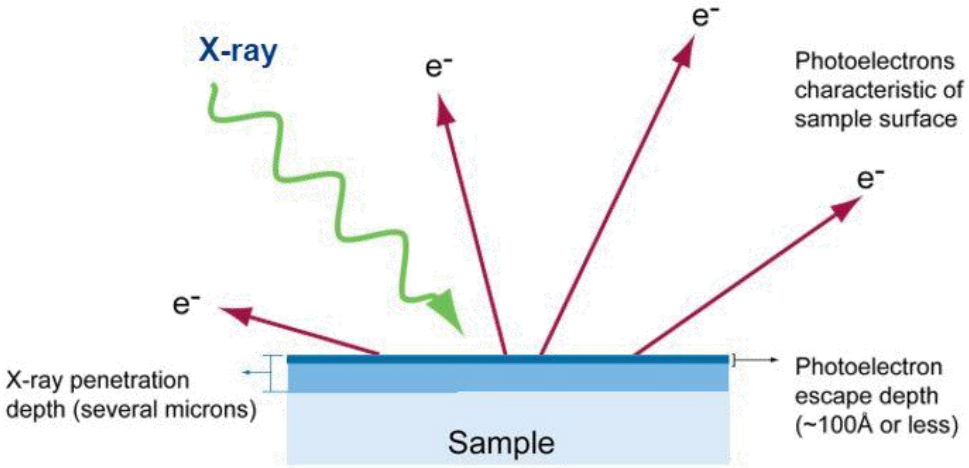


FIG. 1. Photoelectron energy spectra for Ar, Kr, Xe, CS_2, and NO_2.

D.W. Turner and M.I. Al Jobory, J. Chem. Phys. 1962, 37, 3007

X-Ray Photoelectron Spectroscopy (XPS)

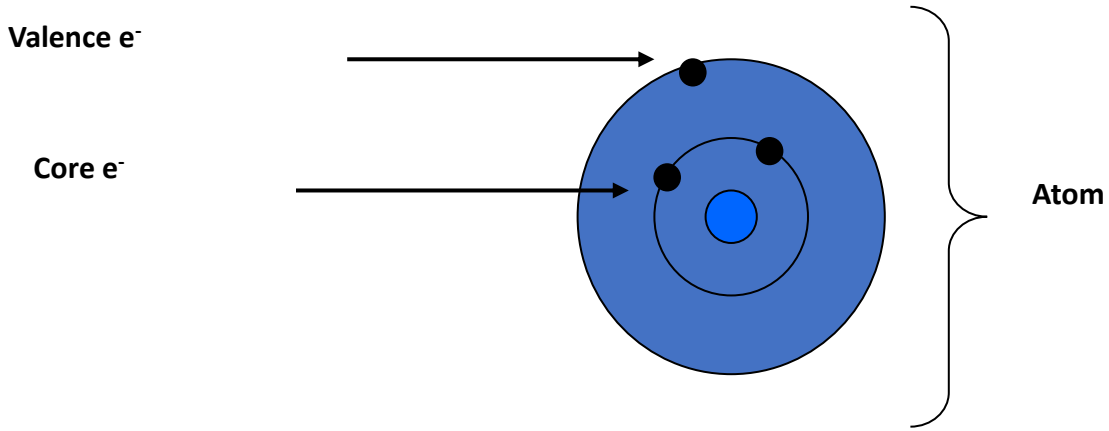
- Irradiate the sample surface, hitting the core electrons (e^-) of the atoms.
- The X-rays penetrate the sample to a depth on the order of a micrometer.
- Useful e^- signal is obtained only from a depth of around 10 to 100 Å on the surface.
- Normally, the sample will be radiated with photons of a single energy.



Why the Core Electrons?

- An electron near the Fermi level is far from the nucleus, moving in different directions all over the place, and will not carry information about any single atom.
  - Fermi level is the highest energy level occupied by an electron in a neutral solid at absolute 0 temperature.
  - Electron binding energy (BE) is calculated with respect to the Fermi level.

- The core e^-s are local close to the nucleus and have binding energies characteristic of their particular element.

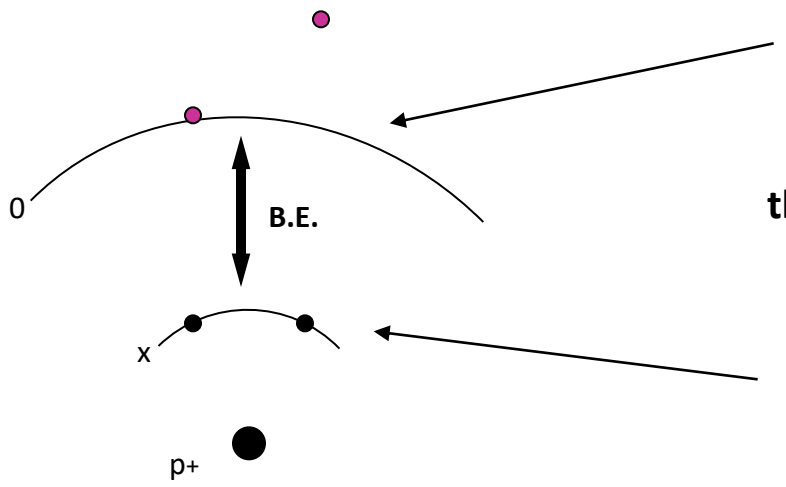


Binding Energy (BE)

The Binding Energy (BE) is characteristic of the core electrons for each element.

The BE is determined by the attraction of the electrons to the nucleus. If an electron with energy x is pulled away from the nucleus, the attraction between the electron and the nucleus decreases and the BE decreases.

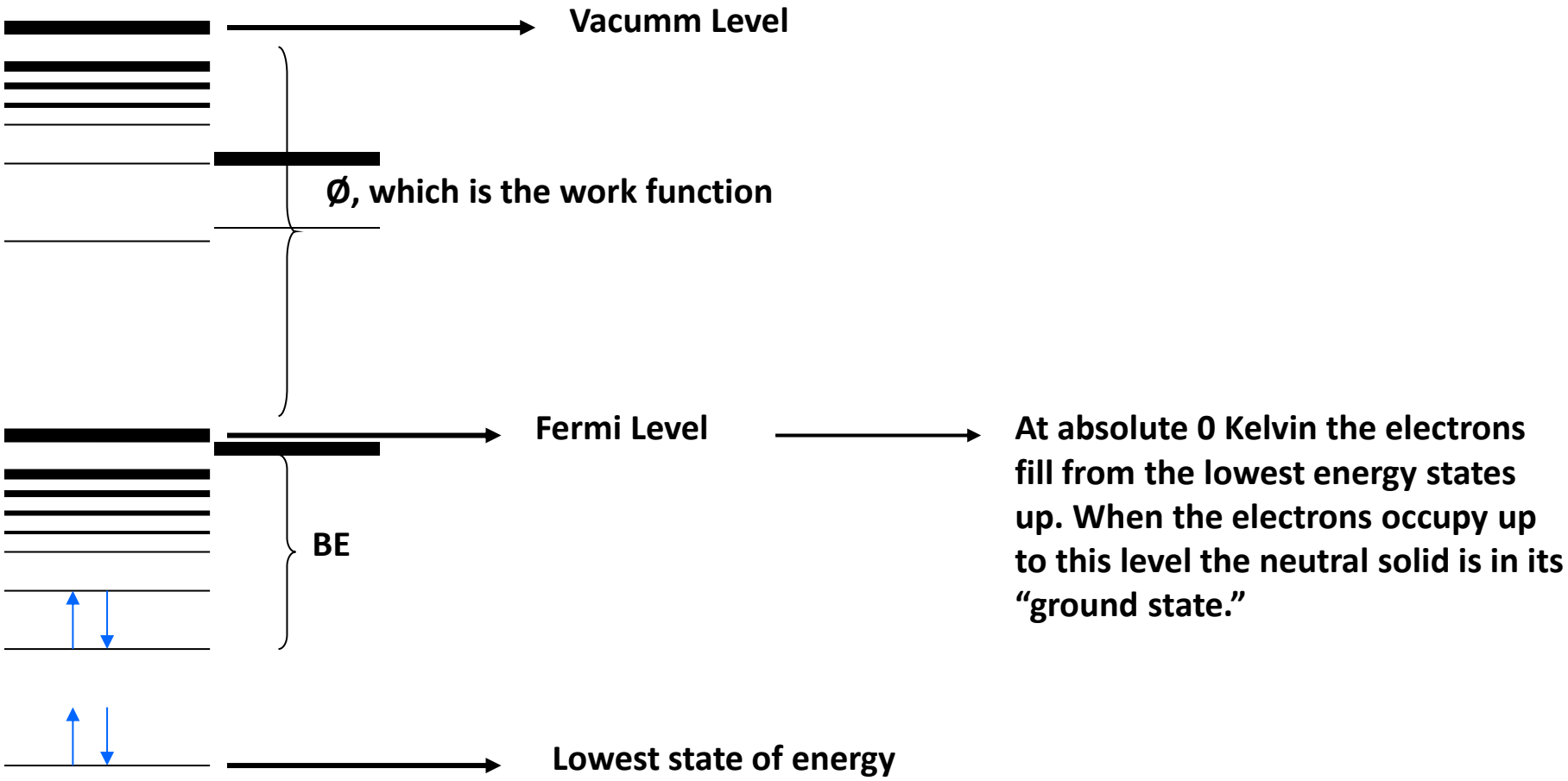
Eventually, there will be a point when the electron will be free of the nucleus.



This is the point with 0 energy of attraction between the electron and the nucleus. At this point the electron is free from the atom.

These electrons are attracted to the proton with certain binding energy x

Energy Levels



Equation

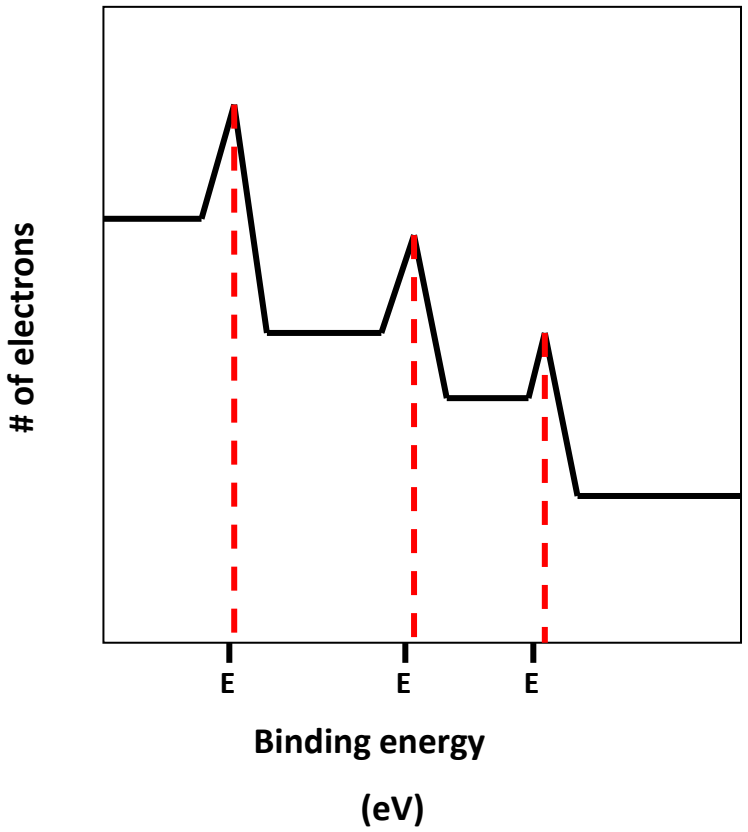
$$KE = h\nu - BE - Ø$$

KE → Kinetic Energy (measure in the XPS spectrometer)
hv → photon energy from the X-Ray source (controlled)
Ø → spectrometer work function. It is a few eV. Found by calibration.

BE → is the unknown variable

The equation will calculate the energy needed to get an e^- out from the surface of the solid.

KE versus BE



KE can be plotted depending on BE

Each peak represents the amount of e^-s at a certain energy that is characteristic of some element.

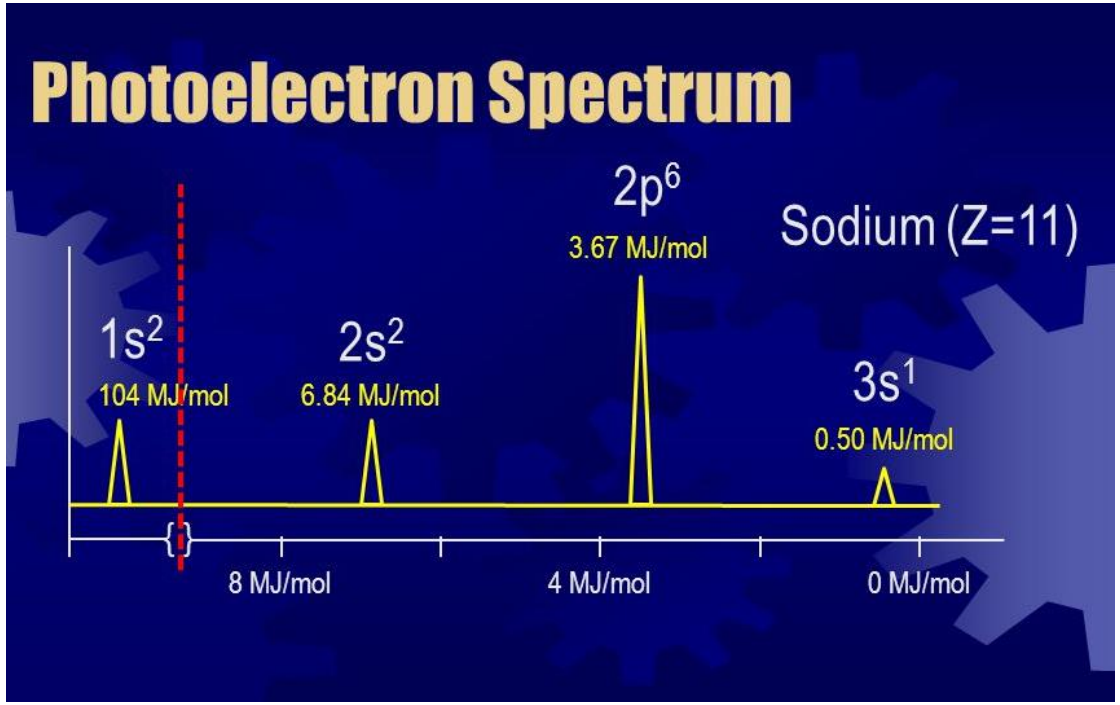
BE increase from right to left

1000 eV ⇄ **0 eV**

KE increase from left to right

Identification of XPS Peaks

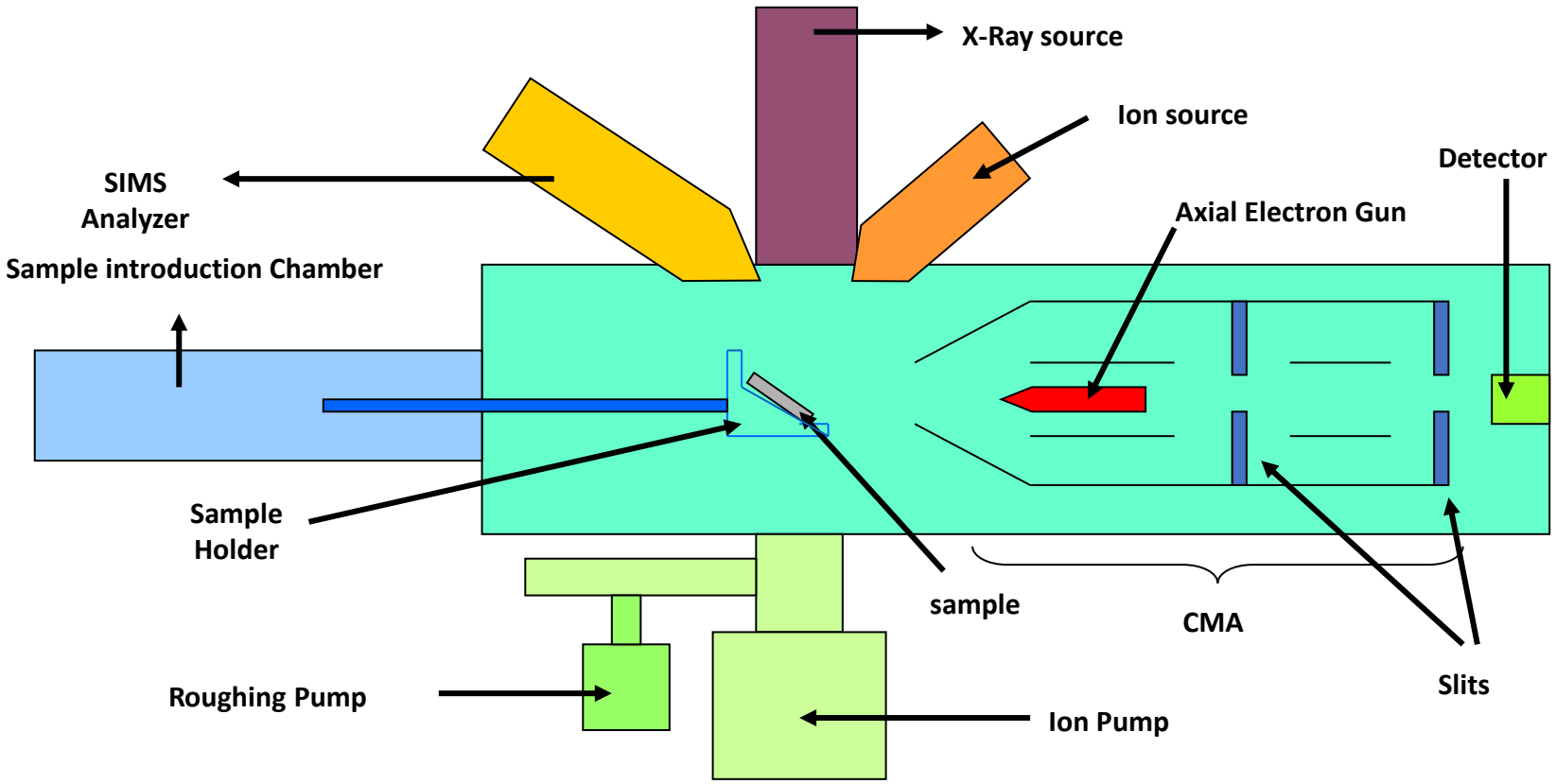
- The plot has characteristic peaks for each element found in the surface of the sample.
- There are tables with the KE and BE already assigned to each element.
- After the spectrum is plotted you can look for the designated value of the peak energy from the graph and find the element present on the surface.



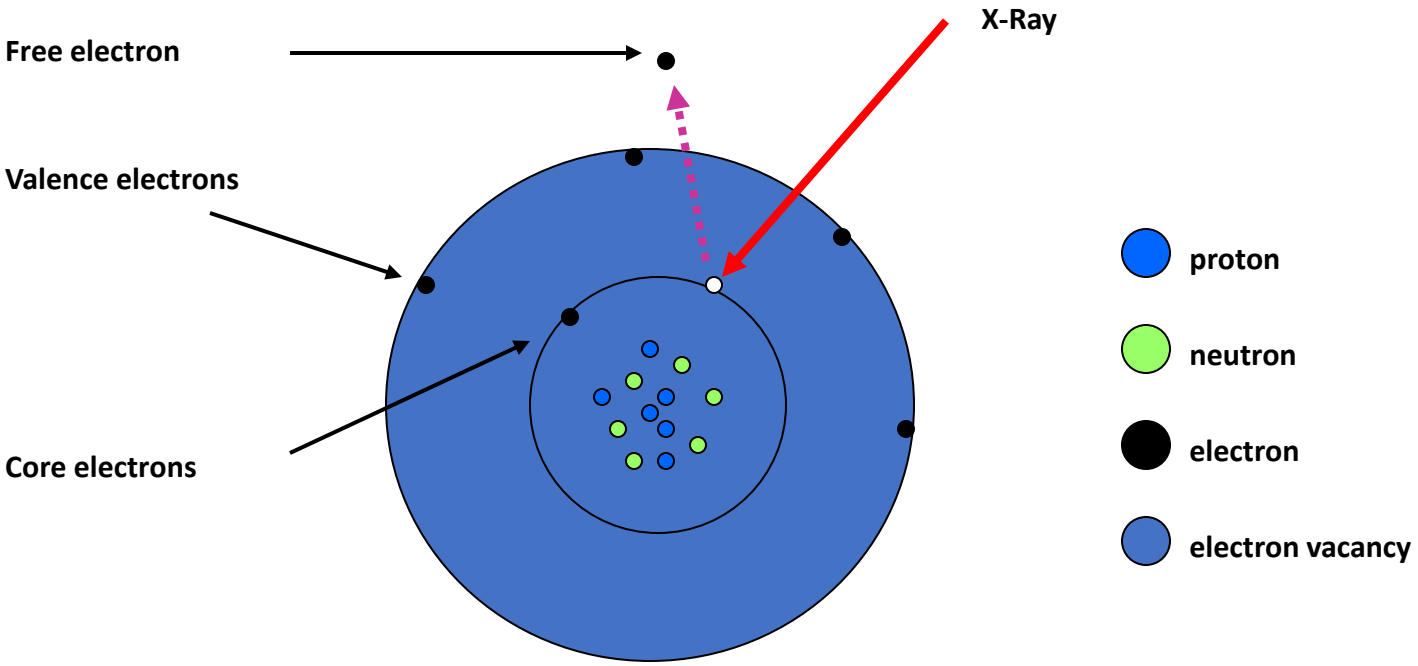
XPS Instrument

- XPS is also known as ESCA (Electron Spectroscopy for Chemical Analysis).
- The technique is widely used because it is very simple to use and the data is easily analyzed.
- XPS works by irradiating atoms of a surface of any solid material with X-Ray photons, causing the ejection of electrons.

Diagram of the Side View of XPS System



The Atom and the X-Ray



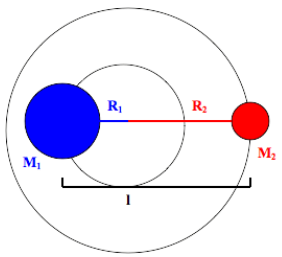
The core electrons respond very well to the X-Ray energy

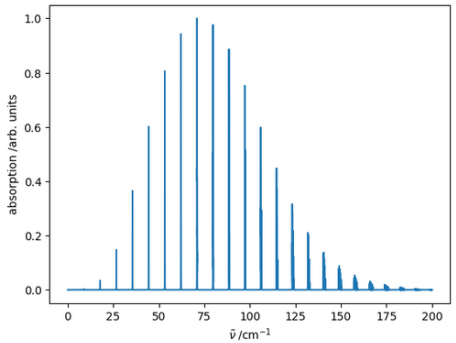
XPS Technology

- Consider as non-destructive
  - because it produces soft x-rays to induce photoelectron emission from the sample surface
- Provide information about surface layers or thin film structures

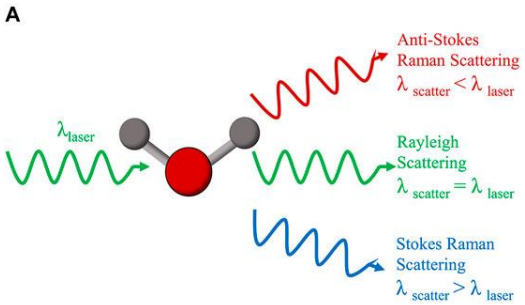
- Applications in the industry:
  - Polymer surface
  - Catalyst
  - Corrosion
  - Adhesion
  - Semiconductors
  - Dielectric materials
  - Electronics packaging
  - Magnetic media
  - Thin film coatings

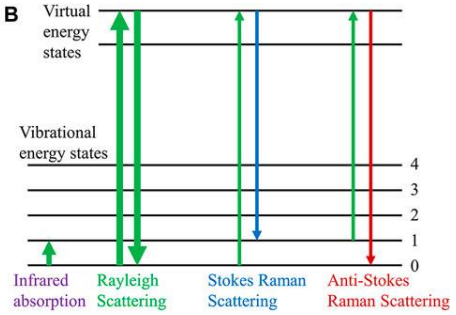
SPECTROSCOPY

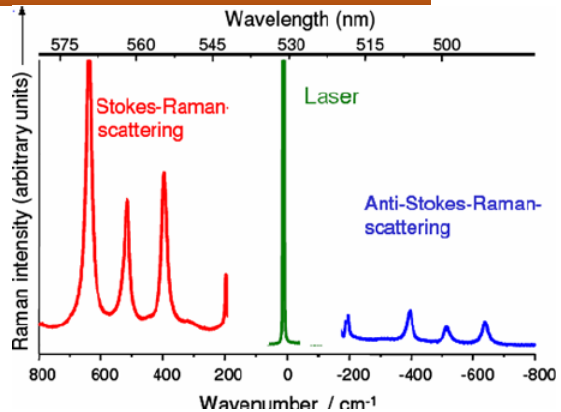




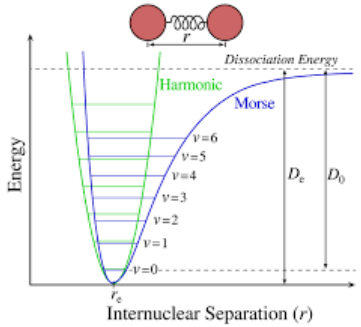
Rotational (MW) Spectroscopy

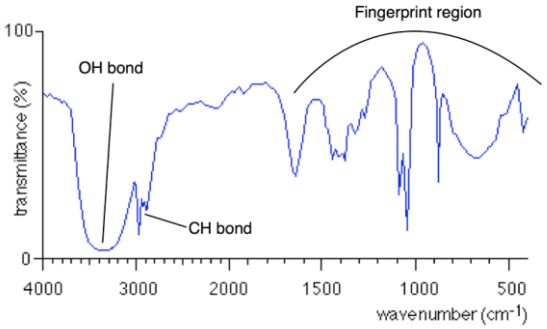




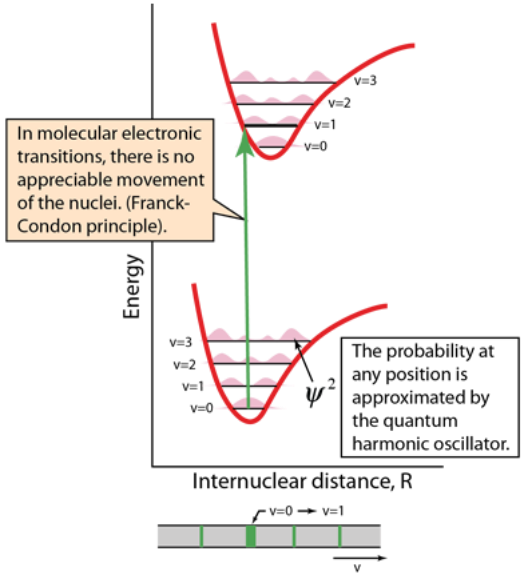


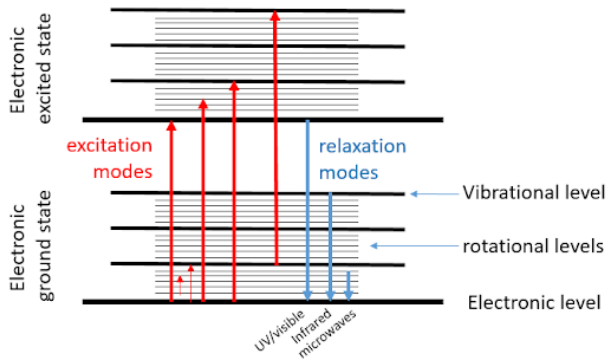
Raman Spectroscopy



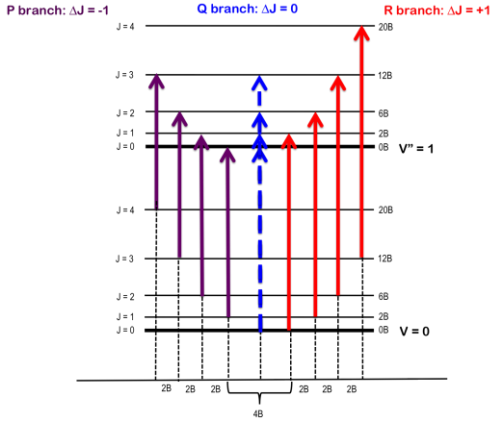


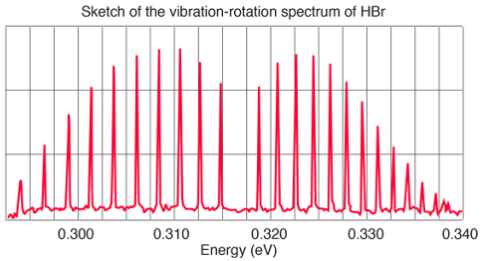
Vibrational (IR) Spectroscopy



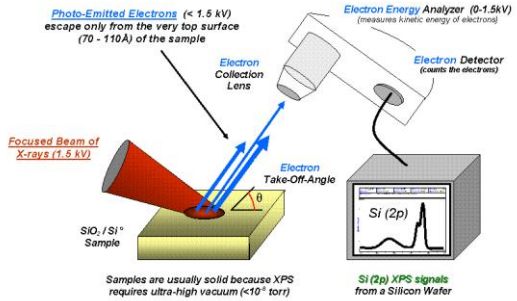


Electronic Spectroscopy





Ro-Vibrational Spectroscopy



Photoelectron Spectroscopy